

cssComm

1 Executive Summary

This module corrects raw Coarse Sun Sensor (CSS) output values to expected cosine values using a pre-calibrated Chebyshev residual model. The residual model maps a normalized raw CSS measurement to the expected deviation from the ideal cosine response. Each sensor measurement is normalized by a configurable maximum sensor value, corrected using the Chebyshev polynomial series, and clamped to the range $[0, 1]$.

2 Message Connection Descriptions

The following table lists all the module input and output messages. The module msg connection is set by the user from python. The msg type contains a link to the message structure definition, while the description provides information on what this message is used for.

Table 1: Module I/O Messages

Msg Variable Name	Msg Type	Description
sensorListInMsg	CSSArraySensorMsgF32Payload	input message containing raw CSS sensor data
cssArrayOutMsg	CSSArraySensorMsgF32Payload	output message of corrected CSS cosine values

3 Module Parameters

The following table lists all the module parameters that can be set. All parameters are required and must be configured before the module is used.

Table 2: Module Parameters

Parameter Name	Type	Units	Default	Description	Bounds
numSensors (required)	uint32_t	[-]	0	Number of CSS sensors in the constellation	Must be in $[1, \text{MAX_NUM_CSS_SENSORS}]$ (checked in setter)
maxSensorValue (required)	float	[-]	0	Maximum raw sensor output value used to normalize measurements	Must be positive (checked in setter)
chebyCount (required)	uint32_t	[-]	0	Number of Chebyshev polynomial coefficients to use	Must be in $[1, 32]$ (checked in setter)
chebyPolynomials (required)	std::array<float, [-32]>		all zeros	Pre-calibrated Chebyshev polynomial coefficients C_i	None

4 Module Assumptions and Limitations

- The Chebyshev residual function is calibrated to a single distance from the sun. As the spacecraft moves farther from the calibration distance, the model loses accuracy. These discrepancies grow increasingly apparent at high sun angles.

- The module assumes `maxSensorValue` is representative of the actual peak sensor output. If this value is incorrect, the normalization and subsequent correction will be inaccurate.
- The Chebyshev correction is applied independently to each sensor. No cross-sensor coupling is modeled.
- The corrected output is clamped to $[0, 1]$. Any sensor reading that produces a corrected value outside this range is saturated.
- The module does not store state between calls. Each `update()` is independent of prior calls.

5 Initialization

The module is configured by:

```
module = cssCommF32.CssComm()
module.modelTag = "cssComm"
module.numSensors = num_sensors
module.maxSensorValue = max_sensor_value
module.chebyCount = len(cheby_list)
module.chebyPolynomials = cheby_list + [0] * (32 - len(cheby_list))
```

The `chebyPolynomials` array must be padded with zeros to fill all 32 entries.

6 Detailed Module Description

6.1 Algorithm Flow

At every update cycle, the `cssComm` module performs the following steps for each of the `numSensors` sensor measurements:

1. **Normalize** the raw sensor reading by `maxSensorValue`:

$$x_{\text{meas}} = \frac{\text{raw}_i}{\text{maxSensorValue}}$$

2. **Compute the Chebyshev correction** δx using the pre-calibrated coefficients:

$$\delta x = \sum_{j=0}^N C_j T_j(x_{\text{meas}})$$

3. **Apply the correction** to the normalized measurement:

$$x_{\text{corr}} = x_{\text{meas}} + \delta x$$

4. **Clamp** the corrected value to the valid cosine range:

$$x_{\text{out}} = \text{clamp}(x_{\text{corr}}, 0, 1)$$

Sensor indices beyond `numSensors` are left at zero in the output message.

6.2 Equations

6.2.1 Residual Function

The correction applied to each CSS measurement is based on a function that maps a raw CSS measurement to the expected cosine response for that measurement (i.e. the distance from the red curve to the blue

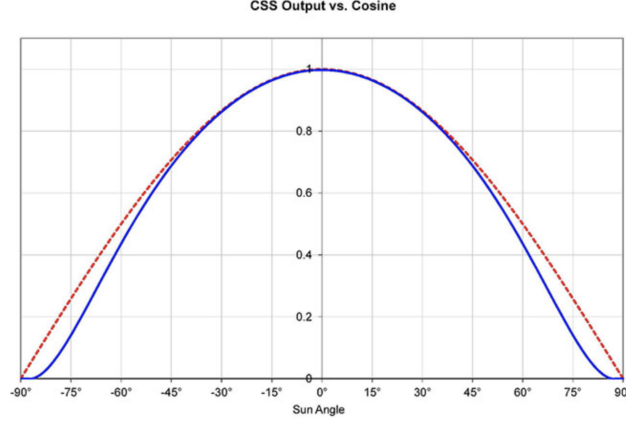


Figure 1: Example of CSS output (blue) relative to a cosine curve (red)

curve in the figure above). This function is modeled using a Chebyshev polynomial series to the N -th order represented by the following form:

$$\delta x = \sum_{i=0}^N C_i T_i(x_{\text{meas}})$$

where $T_i(x)$ represents the Chebyshev polynomials, and C_i are the pre-determined scaling factors.

This correction to the raw measurement is then applied using:

$$x_{\text{corr}} = x_{\text{meas}} + \delta x$$

6.2.2 Chebyshev Polynomial Computation

The procedure to compute the Chebyshev polynomials, $T_i(x)$, is as follows:

1. Suppose we want to evaluate the Chebyshev polynomial of order i at x_0 , denoted $T_i(x_0)$.
2. The first two orders of Chebyshev polynomials are evaluated using the following form:

$$T_0(x) = 1$$

$$T_1(x) = x$$

3. The Chebyshev polynomial of order $i > 1$ can be computed using the values of Chebyshev polynomials of order $i - 1$ and $i - 2$ and the following recursive formula:

$$T_{i+1}(x) = 2xT_i(x) - T_{i-1}(x)$$

4. Apply this formula up to order i to evaluate the Chebyshev polynomial of order i at x_0 .

6.2.3 Output Clamping

After the Chebyshev correction is applied, the result is clamped to the range $[0, 1]$. This ensures that the corrected output represents a valid cosine value. Clamping handles edge cases such as:

- Negative raw sensor readings (e.g. sensor noise) that would produce negative corrected values.
- Sensor readings above `maxSensorValue` whose scaled value exceeds 1.0 after correction.